SKILLSENSE

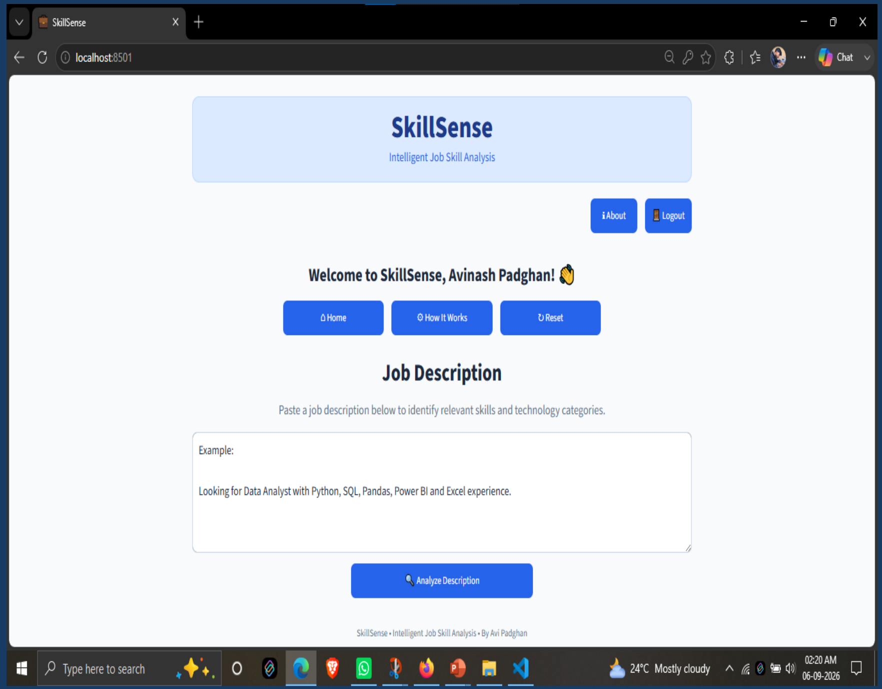
INTELLIGENT JOB SKILL ANALYSIS

An NLP-Based System for Extracting and Analyzing Skills from Job Descriptions

PRESENTED BY: AVINASH PADGHAN

AI • NLP • SKILL INTELLIGENCE

WEB APPLICATION



Live project interface

PROBLEM STATEMENT

Job descriptions contain valuable skill information, but extracting and analyzing these skills manually is time-consuming and inconsistent.

01

Manual Extraction

Identifying required skills from lengthy job descriptions requires significant manual effort.

02

Unstructured Information

Skills are mixed with responsibilities, qualifications, experience, and tools.

03

Difficult at Scale

Analyzing large volumes of job descriptions manually is inefficient.

04

Limited Context

Keyword-based methods can miss related or context-specific skills.

PROJECT NEED → Automate skill extraction, categorization and analysis using NLP.

BUSINESS USE CASE

Turning unstructured job descriptions into actionable skill insights.

01

Recruitment & Hiring

Support faster screening by identifying required skills.

02

Skill Gap Analysis

Compare job requirements with candidate skill profiles.

03

Job Market Analysis

Study skill demand across roles and technologies.

04

Career & Learning

Guide relevant upskilling and learning decisions.

BUSINESS VALUE

Faster Analysis • Better Skill Insights • Data-Driven Decisions

DATASET

TOTAL RECORDS

4,318

job descriptions

DOMAIN
Job & Recruitment

PRIMARY TASK
Skill extraction & analysis

01

DATA CONTENT

- Job descriptions
- Job roles / titles
- Education and experience information
- Required skills and technologies

02

TARGET OUTPUT

Structured skill information extracted from unstructured job-description text for downstream analysis and modeling.

DATA CLEANING

01

Raw Text

Original job descriptions

→

02

Text Cleaning

Remove noise, normalize text and whitespace

→

03

Preprocessing

Prepare text for NLP feature extraction

→

04

Skill Normalization

Standardize skill names and variants

→

05

Clean Dataset

Ready for analysis and modeling

Goal: improve consistency, reduce noise and make the text suitable for NLP models.

EXPLORATORY DATA ANALYSIS

Understanding the dataset before building NLP models.

01

Job Roles

Distribution of job titles and role categories.

02

Education

Comparison of education requirements across records.

03

Skill Demand

Frequency of commonly required skills.

04

Text Characteristics

Length and content patterns in job descriptions.

EDA INSIGHT → These patterns help define useful features, categories and dashboard views.

NLP PIPELINE

01 **Job Description** → 02 **Text Preprocessing** → 03 **Skill Detection** → 04 **Feature Extraction** → 05 **Modeling** → 06 **Skill Insights**

Rule-Based • Machine Learning • Transformer-Based NER • Skill Normalization

RULE-BASED APPROACH

Known-skill matching with dictionary and pattern-based methods.

01

Phrase Matching

spaCy PhraseMatcher detects known skills and skill phrases.

02

Skill Dictionary

A predefined vocabulary identifies technical and non-technical skill mentions.

03

Normalization

Detected variants are mapped to standardized skill names.

SKILL EXTRACTION PERFORMANCE

	Method	Accuracy	Precision	Recall
Dictionary	0.620	0.583	0.620	0.601
Regex	0.052	0.567	0.052	0.096

Measured on the final skill extraction comparison.

MACHINE LEARNING APPROACH

TF-IDF + Logistic Regression for skill-category classification.

JOB TEXT → TF-IDF → LOGISTIC REGRESSION → SKILL CATEGORY

MODEL RESULT

Accuracy **0.2143**

14 test samples

CLASSIFICATION REPORT — KEY RESULT

SKILL class: precision 0.21 • recall 1.00 • F1 0.35

Macro avg F1: 0.04 • Weighted avg F1: 0.08

TRANSFORMER APPROACH

Entity-level skill extraction using a DistilBERT NER pipeline.

01

MODEL

DistilBERT-based NER pipeline using dslim/distilbert-NER.

02

EXTRACTION

Token classification identifies entities from job-description text; aggregation groups token pieces.

03

DEPLOYMENT READY

The model can be stored locally with configuration, weights and tokenizer files.

SKILL EXTRACTION PERFORMANCE

Transformer **Accuracy 0.636 | Precision 0.916 | Recall 0.636 | F1 0.751**

TOKEN OUTPUT EXAMPLE

del → ORG 0.943
##oit → ORG 0.979
##te → ORG 0.984

FINAL SKILL EXTRACTION COMPARISON

Performance across four skill extraction methods.

Method	Accuracy	Precision	Recall	F1 Score
Dictionary	0.620	0.583	0.620	0.601
Regex	0.052	0.567	0.052	0.096
Semantic Similarity	0.036	0.852	0.036	0.069
Transformer	0.636	0.916	0.636	0.751

BEST OVERALL EXTRACTION RESULT → Transformer: Accuracy 0.636 • Precision 0.916 • Recall 0.636 • F1 0.751

ERROR ANALYSIS

01

Ambiguous Terms

A term may have different meanings depending on job context.

02

Unseen Skills

Skills not represented in training vocabulary or patterns can be missed.

03

Tokenization

Subword tokenization can split skill names across multiple tokens.

04

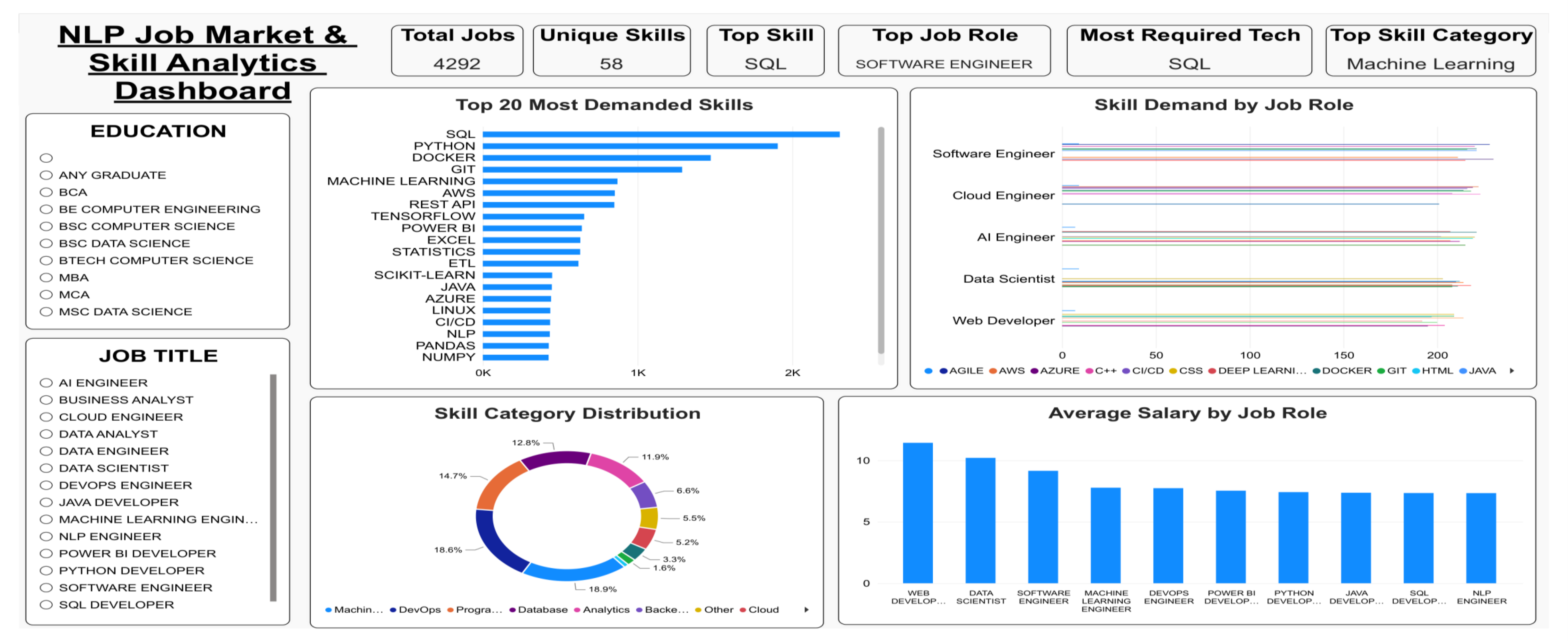
Similar Entities

Related technologies or phrases can be difficult to separate precisely.

Observed transformer tokenization example: del / ##oit / ##te were classified as ORG.

DASHBOARD

NLP Job Market & Skill Analytics — Power BI Dashboard

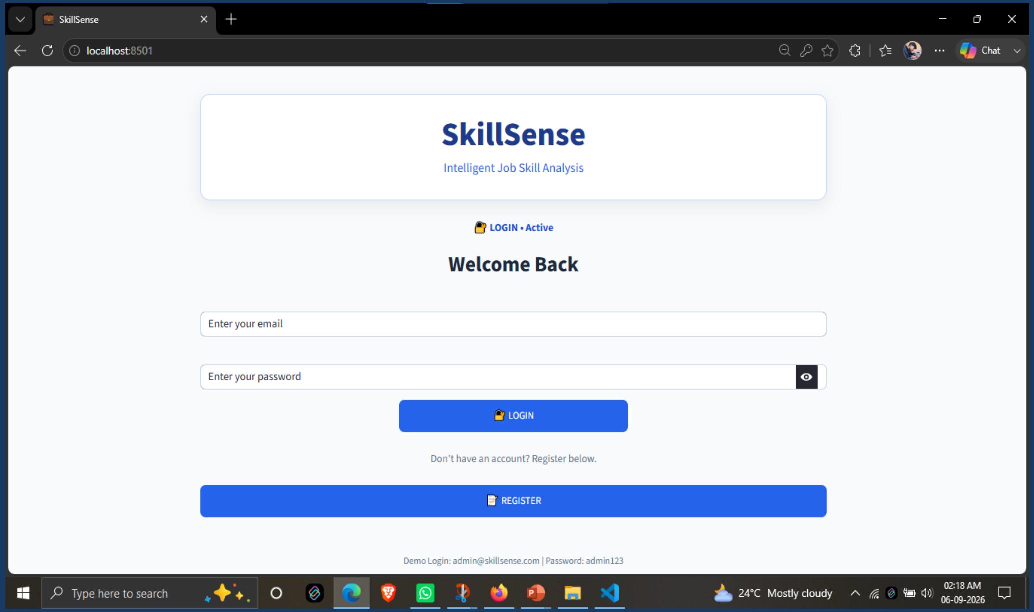


Dashboard highlights: 4,292 total jobs • 58 unique skills • Top skill: SQL • Top job role: Software Engineer • Top category: Machine Learning

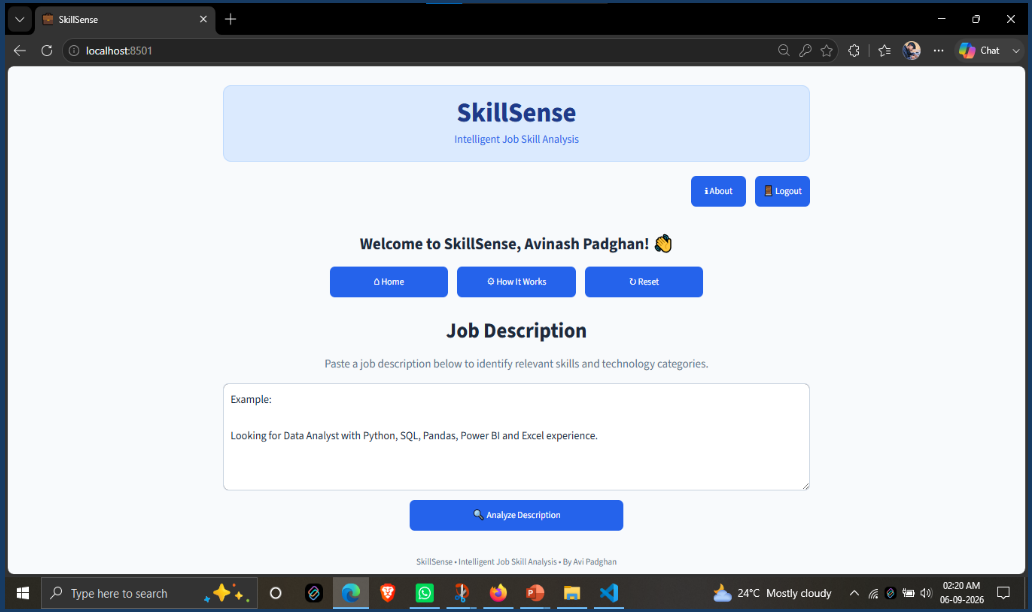
APPLICATION DEMO

SkillSense web application for intelligent job-description analysis.

01 LOGIN



02 ANALYSIS HOME



LOGIN → ENTER JOB DESCRIPTION → ANALYZE → VIEW SKILL RESULTS

CONCLUSION

01

Automated Analysis

NLP can reduce the manual effort required to identify skills in job descriptions.

02

Multiple Approaches

Rule-based, classical ML and transformer methods provide complementary capabilities.

03

Actionable Insights

Extracted skills can support recruitment, market analysis and career decisions.

SkillSense bridges unstructured job text and structured skill intelligence.

FUTURE IMPROVEMENTS

01

Domain Fine-Tuning

Fine-tune transformer models on domain-specific job-description data.

02

Skill Ontology

Build a richer skill taxonomy with related skills, aliases and technology families.

03

Advanced Matching

Add resume-to-job skill matching and automated skill-gap scoring.

04

Live Market Insights

Integrate updated job data to monitor changing skill demand over time.

THANK YOU